

Title: Strength of Materials

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Chapter 6: Torsion of Shafts

Torsion is a classic problem in mechanical design, arising whenever a shaft transmits power through rotation.

Example 6.1 illustrates a common misconception about shaft stiffness and is discussed qualitatively only; no numerical solution is given here.

Example 6.2: A solid circular shaft of diameter 50 mm transmits a torque of 2 kN*m. Determine the maximum shear stress $\tau_{\max}$ in the shaft, given that $\tau_{\max} = T r / J$.

Problem 6.5: A hollow circular shaft has outer diameter 60 mm and inner diameter 40 mm, and carries a torque of 3 kN*m. Find the shear stress at the outer surface.

Chapter 7: Fluid Mechanics